

Set A

Student ID:	Name:	Section:
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[Answer any **THREE** questions out of **FOUR**. Each question carries **15 marks**. Return the question paper with the answer script.]

1. (CO4) An initial cut for a partition of 8 circuit elements (a, b, c, d, e, f, g, and h) is given in **Figure 1**.

The cut line divides the circuit elements into two partitions named **A** and **B**.

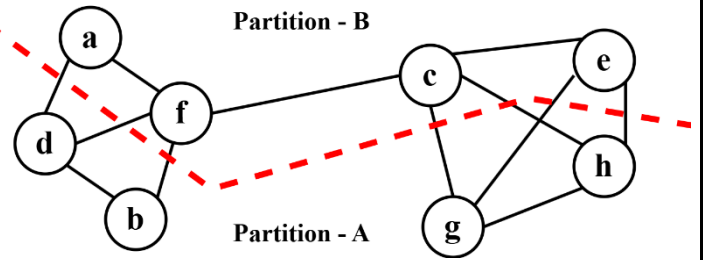


Figure 1

- (a) Calculate the initial cut cost, and the cost of moving each node. [1]

- (b) Perform the first pass of the algorithm. For every “i”th iteration inside the first pass, until all the nodes are swapped and fixed, do the following: [9]
- Compute the node costs of all unfixed nodes
 - Find the maximum gain of swapping a pair of nodes (Δg_i)
 - Swap the pair and draw the updated graph

- (c) Compute G_i of each iteration and find the maximum $G_i = G_m$. Suggest how many swaps should be actually executed in the first pass. Redraw the optimized graph. [4]

- (d) Should you perform subsequent passes of the algorithm? Why or why not? [1]

2. (CO1) Brook is a chip designer who works specifically on the $\lambda = 1.5 \mu\text{m}$ process. However, he has messed up the layout design of a 3-input NOR gate (Figure 2(a)) because of which, the NOR gate has **an equivalent rise resistance that is four times its equivalent fall resistance**. Also, the NOR gate's **equivalent rise resistance is twice the rise resistance of a unit inverter** for that design process. But at least he made sure to keep the source-drain terminals shared where possible to reduce capacitance. The gate capacitance and parasitic (diffusion) capacitance of are $0.5 \text{ fF}/\mu\text{m}$.

This **NOR gate** is loaded with a **Unit inverter** and the output node Y has a voltage waveform as shown in Fig. 2(c).

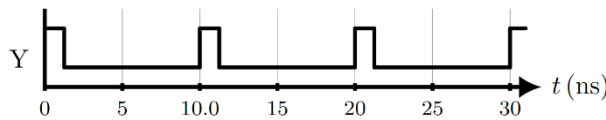


Figure 2(c): Voltage waveform at node Y

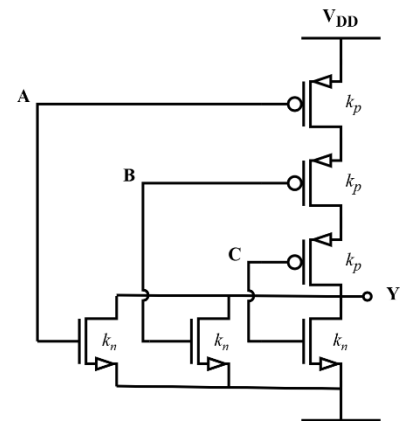


Figure 2(a): CMOS implementation of a 3 - input NOR gate

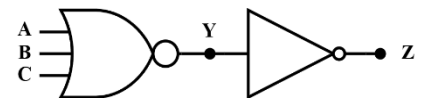


Figure 2(b): A 3-input NOR gate loaded with a Unit Inverter

- (a) Find the value of width scaling factors - k_p and k_n of the NOR gate designed by Brook. Thereafter, calculate the actual width of the pMOS and nMOS. [2]

- (b) Draw the RC equivalent model of the above 3-input NOR gate. Lump the capacitances together where possible. [5]

- (c) Find the capacitance of the Y node (in fF units), when it is loaded with a unit inverter, as shown in Fig. 2(b). [3]

- (d) Find the best-case falling edge contamination delay (t_{cdf}) of Y-node in terms of R and C of unit MOS. [2]
- [Hint: You do not need Elmore modelling. All three PMOS are off. Three NMOS resistances are in parallel.]

- (e) Find the switching power of the Y-node, if the system has a supply voltage of 5 V. [3]

3. (CO4) For question (a) you are given a certain **chip planning task**. You have three blocks with the following widths and heights:

Block A: $w = 2, h = 3$ or $w = 3, h = 2$

Block B: $w = 1, h = 2$ or $w = 2, h = 1$

Block C: $w = 1, h = 4$ or $w = 4, h = 1$

For questions (b-e) refer to the adjacent figures on the right of the paper, you can see the grid layout of a system with a source (S) and two targets – T1 and T2. The obstacles and free cells are shown.

(a) **Define** chip planning. **Find** and draw a floorplan with the minimum [3] total area enclosed by its global bounding box.

(b) Using the Lee's Maze Algorithm, **find** the shortest path from S to T1 [4] and T2 *allowing minimum bends*.

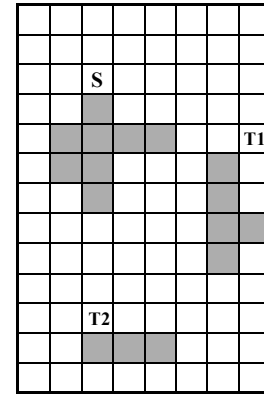
Answer this question on the grids on the question paper.

(c) **Find** the total memory usage of the given grid if the sequence is a [3] series of natural numbers.

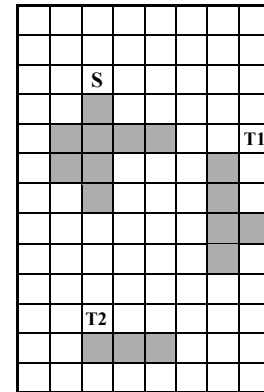
(d) If the maximum memory requirement of the provided grid is **36 bytes**, [3] **calculate** the maximum number of bits that each cell can hold.

(e) **Find** the number of consecutive and unique numbers (starting from 1) [2] that can be used for wave propagation in the Lee's Maze algorithm, for the given memory constraint.

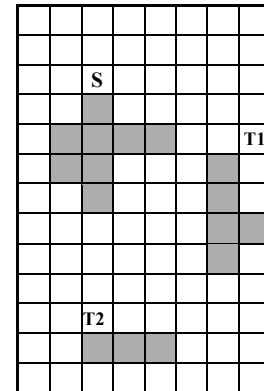
What should be a full sequence for wave propagation?



Obstacle
Free Cell



Obstacle
Free Cell



Obstacle
Free Cell

4. (CO1) Suppose the technology you are using to design a VLSI system has $\lambda = 80$ nm, a clock frequency of **5 MHz**, and a supply is **5 V**. The chip you are designing has **5 million transistors**, of which 1 million remain **active** at any given time. The activity factor is defined by the fraction of total components which remain active. The gate and diffusion capacitances are $12 \text{ fF}/\mu\text{m}$ and $5 \text{ fF}/\mu\text{m}$, respectively for all the 5 million transistors. The gate width is **20 λ** . You also obtain the following power consumption data:

- Short circuit power = 0.5 W
- Leakage power = 0.01 W
- Subthreshold power = 0.02 W

The **acceptable TOTAL power consumption** of a chip is **3 W**.

(a) **Find** the activity factor and load capacitance of the system described above. [2]

(b) **Calculate** the switching power consumption of the chip. [3]

(c) **Calculate** the dynamic and static power of the chip. Thereafter calculate the **TOTAL** power consumption. [3]

(d) Is the **TOTAL** power consumption within the acceptable range? If not, **find** the maximum clock frequency to keep the **TOTAL** power within the acceptable range? [4]

(e) For a **10-input NAND gate**, what should be the ratio of width scaling factors – k_p/k_n - for the NMOS and PMOS [3] to keep the rise and fall resistances equal?

Set B

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[Answer any **THREE** questions out of **FOUR**. Each question carries **15 marks**. Return the question paper with the answer script.]

- 1. (CO4)** An initial cut for a partition of 8 circuit elements (**a, b, c, d, e, f, g, and h**) is given in **Figure 1**.

The cut line divides the circuit elements into two partitions named **A** and **B**.

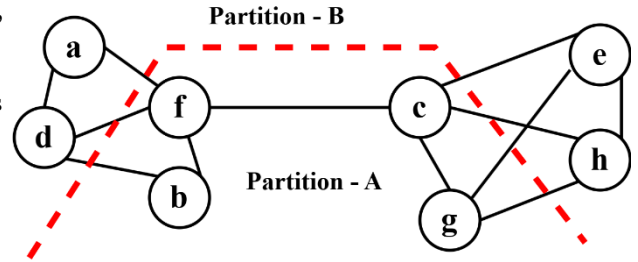


Figure 1

- (a) Calculate the initial cut cost, and the cost of moving each node. [1]
- (b) Perform the first pass of the algorithm. For every “i”th iteration inside the first pass, until all the nodes are swapped and fixed, do the following: [9]
- Compute the node costs of all unfixed nodes
 - Find the maximum gain of swapping a pair of nodes (Δg_i)
 - Swap the pair and draw the updated graph
- (c) Compute G_i of each iteration and find the maximum $G_i = G_m$. Suggest how many swaps should be actually executed in the first pass. Redraw the optimized graph. [4]
- (d) Should you perform subsequent passes of the algorithm? Why or why not? [1]

- 2. (CO1)** Bob is a chip designer who works specifically on the $\lambda = 0.25 \mu\text{m}$ process. However, he has messed up the layout design of a **3-input NAND gate** (Figure 2(a)) because of which, the NAND gate has an equivalent rise resistance that is four times its equivalent fall resistance. Also, the NAND gate’s equivalent rise resistance is 1.5 the rise resistance of a unit inverter for that design process. But at least he made sure to keep the source-drain terminals shared where possible to reduce capacitance. The gate capacitance and parasitic (diffusion) capacitance of are **2.42 fF/ μm** .

This **NOR gate** is loaded with a **Unit inverter** and the output node **Y** has a voltage waveform as shown in Fig. 2(c).

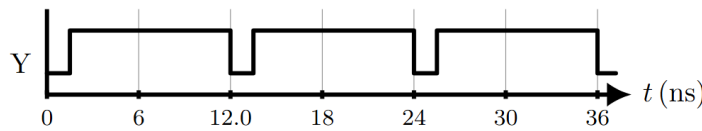


Figure 2(c): Voltage waveform at node Y

- (a) Find the value of width scaling factors - k_p and k_n of the NOR gate designed by Bob. Thereafter, calculate the actual width of the **pMOS** and **nMOS**. [2]

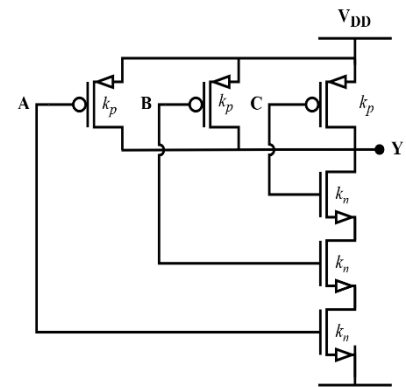


Figure 2(a): CMOS implementation of a 3 - input NAND gate

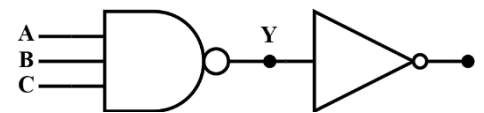


Figure 2(b): A 3-input NAND gate loaded with a Unit Inverter

- (b) Draw the RC equivalent model of the above 3-input NAND gate. Lump the capacitances together where possible. [5]
- (c) Find the capacitance of the **Y** node (in **fF** units), when it is loaded with a unit inverter, as shown in Fig. 2(b). [3]
- (d) Find the best-case rising edge **contamination delay** (t_{cdr}) of **Y**-node in terms of **R** and **C** of unit MOS. [2]
- [Hint: You do not need Elmore modelling. All three NMOS are off. Three PMOS resistances are in parallel.]
- (e) Find the switching power of the **Y**-node, if the system has a supply voltage of **3.3 V**. [3]

3. (CO4) For question (a) you are given a certain **chip planning task**. You have three blocks with the following widths and heights:

Block A: $w = 1, h = 4$ or $w = 4, h = 1$

Block B: $w = 1, h = 2$ or $w = 2, h = 1$

Block C: $w = 2, h = 3$ or $w = 3, h = 2$

For questions (b-e) refer to the adjacent figures on the right of the paper, you can see the grid layout of a system with a source (S) and two targets – T1 and T2. The obstacles and free cells are shown.

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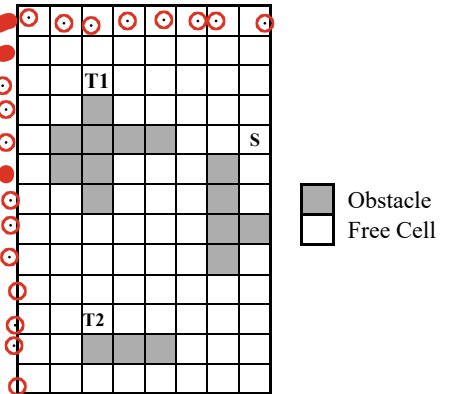
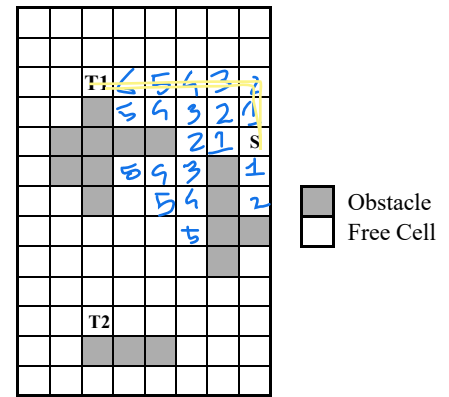
Answer this question on the grids on the question paper.

(c) **Find** the total memory usage of the given grid if the sequence is a [3] series of natural numbers.

(d) If the maximum memory requirement of the provided grid is **24 bytes**, [3] **calculate** the maximum number of bits that each cell can hold.

(e) **Find** the number of consecutive and unique numbers (starting from 1) [2] that can be used for wave propagation in the Lee's Maze algorithm, for the given memory constraint.

What should be a full sequence for wave propagation?



4. (CO1) Suppose the technology you are using to design a VLSI system has $\lambda = 40$ nm, a clock frequency of **2.5 MHz**, and a supply is **5 V**. The chip you are designing has **10 million transistors**, of which **4 million** remain **active** at any given time. The activity factor is defined by the fraction of total components which remain active. The gate and diffusion capacitances are $12 \text{ fF}/\mu\text{m}$ and $5 \text{ fF}/\mu\text{m}$, respectively for all the 10 million transistors. The gate width is **20 λ** . You also obtain the following power consumption data:

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The **acceptable TOTAL power consumption** of a chip is **3 W**.

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(e) For a **10-input NAND gate**, what should be the ratio of width scaling factors – k_p/k_n - for the NMOS and PMOS [3] to keep the rise and fall resistances equal?